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WHAT LESSONS CAN WE LEARN TO KEEP EUROPEAN INDUSTRY COMPETITIVE FROM THE MEASURES TAKEN BY GERMANY, AS THE COST OF ELECTRICITY CONTINUES TO TURN THE SCREW?

Marianne Mueller and Pascal Devaux

The 2022 energy crisis clearly highlighted the issue of electricity prices in Europe – and, more broadly, the issue of energy prices. As a result, European manufacturers' competitiveness has been severely undermined when compared with the United States and China. By reigniting tensions over electricity prices, the conflict that began in the Middle East in late February 2026 further underscores the continent's vulnerability to fossil fuel imports, despite the expansion of renewable energy. Faced with an environment of potential sharp and sudden price rises, European governments do not have the same room for manoeuvre.

While the additional costs borne by European manufacturers are mainly linked to the share of gas in national electricity mixes, we will first illustrate that, in Germany, grid operation and maintenance costs, as well as taxation, are also contributing significantly to these costs.

We will then explore the solutions proposed by the German government under the new European measures regulating state aid to electricity-intensive industries.

Finally, our estimates will show that, while these measures make Germany more competitive against some European countries, they do not place it on an equal competitive footing with the United States or China. Nevertheless, Germany has made significant progress in decarbonising and making its electricity mix more flexible from a European perspective. This should help to ensure that electricity costs are reduced in the medium term.

ECONOMIC RESEARCH

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GAS: A KEY FACTOR IN THE ADDITIONAL ELECTRICITY COSTS FACED BY EUROPEAN INDUSTRY

European manufacturers are still paying significantly higher electricity costs than their American and Chinese competitors. In 2019, there was a 30% difference between the prices charged to European electricity-intensive industries (corresponding to the lower end of the industrial price range) and the average US industrial price. In 2025, it stood at nearly 100% according to the IEA (2026)[1], after peaking at an average of 148% between 2022 and 2023. There is currently a 50% difference with their Chinese competitors[2].

The disruption to Russian gas supplies since 2022 is behind the rising gas prices, particularly as the EU has had to turn to liquefied natural gas[3] (LNG), which is more expensive than gas transported via pipeline and, above all, exposes Europe to global gas market conditions. The growth of LNG has, in fact, intensified competition between the major consumption regions, primarily Europe and Asia. European and Asian prices now tend to move more in tandem (*Chart 1*). However, in China, the prevalence of coal — a widely available local resource — and a more regulated electricity price help to keep industrial prices lower. At the same time, the United States enjoys plentiful domestic gas, which is produced at a low cost, driven by the sharp rise in shale oil and gas production over the past decade. By contrast, since 2022, Europe has been more exposed to the cost of imported gas (90% of its total consumption, according to Eurostat), while the United States and China have enjoyed plentiful domestic resources.

THE 'MERIT ORDER' PRINCIPLE IN SETTING ELECTRICITY PRICES IN EUROPE

Fluctuations and the level of gas prices are passed on to European electricity prices via the 'merit order' marginal pricing mechanism, with the wholesale price determined by the marginal cost of the last (and most expensive) electricity source called upon to meet demand.

As a result, gas plays a decisive role, with gas-fired power stations setting the price in around 40% of cases in Europe[4], with significant variations depending on national electricity mixes. It is close to this average in Germany, below it in Spain (26%) and France (17%), and above it in Italy (73%)[5] in 2025[6] [7]. Therefore, electricity prices are still sensitive to fluctuations in gas prices, even when electricity is not generated from gas (*Chart 2*). This pass-through is particularly pronounced when demand rises and requires the use of more expensive sources, or when supply constraints arise (for example, during extreme heatwaves that reduce wind-power generation or disrupt the cooling of nuclear power plants).

The price of US gas: a decisive advantage

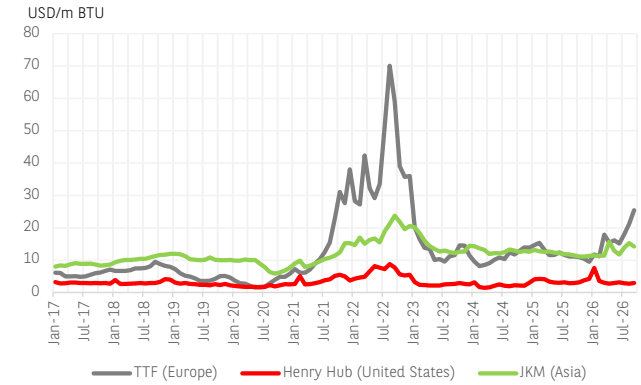


CHART 1 SOURCE: CEIC, BNP PARIBAS

There is still a major link between gas prices and wholesale electricity prices in Europe

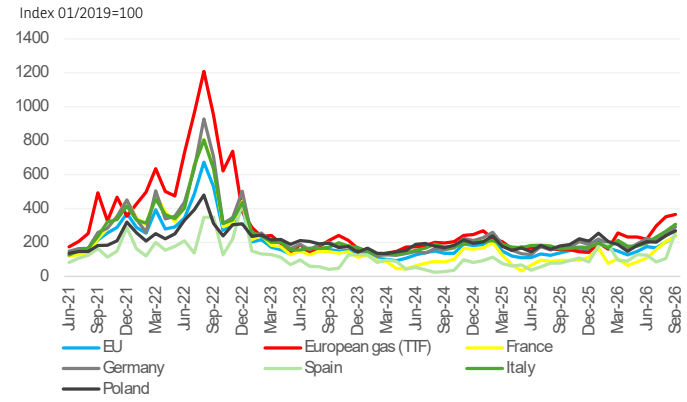


CHART 2 SOURCE: EMBER, BLOOMBERG, BNP PARIBAS

However, for the time being, the 2026 energy crisis is having a far smaller effect on electricity prices than in 2022. On the one hand, the 2026 peak reached by the European TTF benchmark is still well below that recorded in 2022[8].

In addition, there has been significant progress in decarbonising European electricity mixes since 2022, which has helped to mitigate, on average, the pass-through of gas prices to electricity prices.

1 Prices – Electricity 2026 – Analysis – IEA
2 The difference may be even larger for other manufacturers who do not receive the same support as electricity-intensive sectors.
3 Imports of Russian gas (via pipeline and LNG) accounted for around 45% of total European imports in 2021, compared with 14% in Q2 2026 (Bruegel, natural gas imports).
4 ACER, 2026, Key developments in EU electricity and gas markets, 2026 Monitoring report.
5 In Italy, in response to sharply rising gas prices in 2026, the government is currently negotiating a potential option of capping the prices of gas supplied to power stations in order to limit the rise in the wholesale price of electricity. However, this can only be a temporary general mechanism, with the potentially major cost being borne by public funds and/or households.
6 Eurelectric, 2026, *Power Barometer*.
7 EMBER, 2026, Latest energy shock reminds Europe of its risky reliance on gas.
8 Energy shock: 2026 vs. 2022 dashboard – 2nd October 2026 – Economic Studies – BNP Paribas

Europe is diversifying its electricity mixes as part of its decarbonisation agenda

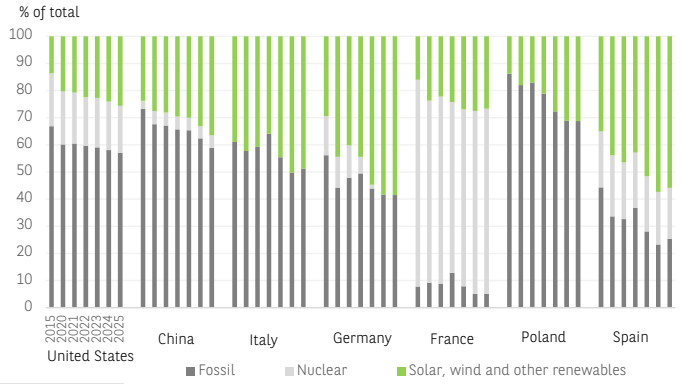
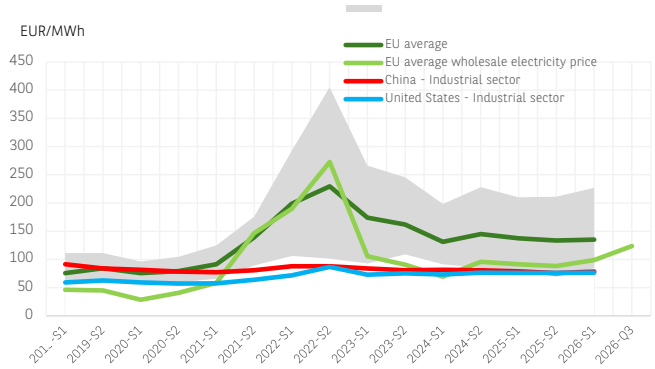


CHART 3

SOURCE: EMBER, BNP PARIBAS

Electricity prices for energy-intensive industries: differences in competitiveness that still need to be qualified

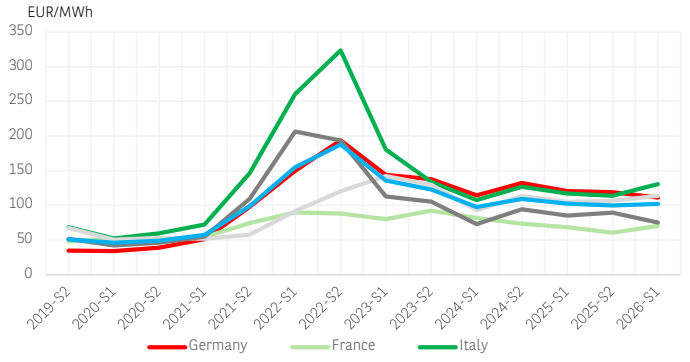


European prices are for industries with consumption > 150k MWh, all taxes and levies included

CHART 5

SOURCE: EUROSTAT, EIA, CEIC, BNP PARIBAS

There are major differences in electricity prices for electricity-intensive industries within Europe



European prices are for non household with consumption > 150k MWh, excl. taxes and levies

CHART 4

SOURCE: EUROSTAT, BNP PARIBAS

Breakdown of electricity prices for European manufacturers

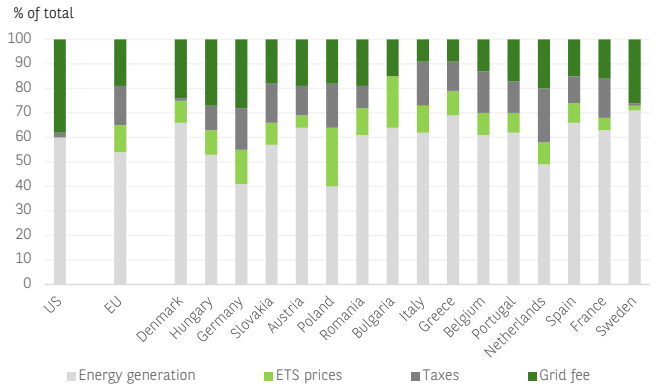


CHART 6

SOURCE: EUROPEAN COMMISSION, BNP PARIBAS

SOME EUROPEAN COUNTRIES HAVE MANAGED TO STAY COMPETITIVE, BUT GERMANY IS STILL THE EXCEPTION

Electricity prices partly vary due to countries' electricity mixes (*Chart 3*). Electricity-intensive manufacturers in countries where the share of fossil energy is high face higher costs (Poland, Italy and Ireland) (*Chart 4*). Conversely, countries with a less carbon-intensive mix (particularly France and Spain in recent times) are less directly exposed to these fluctuations, with costs closer to the United States and China (*Chart 5*).

The situation in Germany is paradoxical, as, despite progressively greening its electricity mix and wholesale electricity prices (averaging 90 EUR/MWh over the 2023–2025 period) and average electricity costs (across all industrial sectors) in line with the European average, the most energy-intensive industries face higher costs than the European average (and even more so compared with international competitors). This is mainly due to grid costs and a tax burden above EU standards (*Chart 6*).

In the first half of 2026, for the most electricity-intensive industries, the difference compared with the European average was 9%, based on the prices excluding taxes and levies provided by Eurostat. This gap is mainly due to the cost of accessing the grid. If all taxes and levies are included, the difference rises to 21% (*Chart 7*)[9].

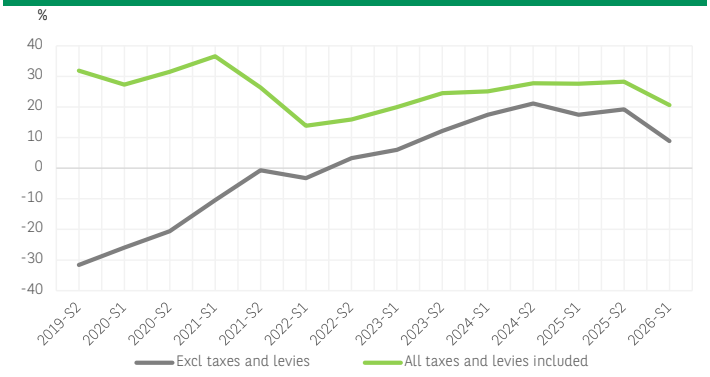
EUROPE'S APPROACH OF ADOPTING BOTH STRUCTURAL PROGRAMMES AND TARGETED SUPPORT MEASURES

European programmes to support the low-carbon transition pursue objectives relating to climate action (reducing greenhouse gas emissions), sovereignty[10] (increasing energy self-sufficiency) and competitiveness (reducing the volatility and costs associated with fossil fuels). The RePowerEU programme, which was launched in 2022 in response to the energy crisis, primarily aims to reduce imports of Russian gas and, more generally, to accelerate the low-carbon transition. Although Europe has reduced its dependence on imports of Russian fossil fuels, there has still not been enough progress around the energy transition and cross-border connectivity. This is due to governance constraints

9 Eurostat data are used to compare prices with the rest of the EU. However, given that electricity prices are not uniform for businesses at a European level, there may be differences between Eurostat and national sources. Jäger P., 2025, "*Cautionary tale or model for Europe? How member states should react to Germany's new electricity subsidy*", Hertie School, Jacques Delors Centre.

10 P. Devaux, 2026, "European Union: low carbon transition and energy sovereignty, a path fraught with obstacles"

A system of taxes and levies that harms Germany's electricity-intensive industries



Electro-intensive industry electricity prices gap between Germany and average EU (>150k MWh)

CHART 7 SOURCE: EUROSTAT, BNP PARIBAS

Germany: sharp decline in output from energy-intensive industries

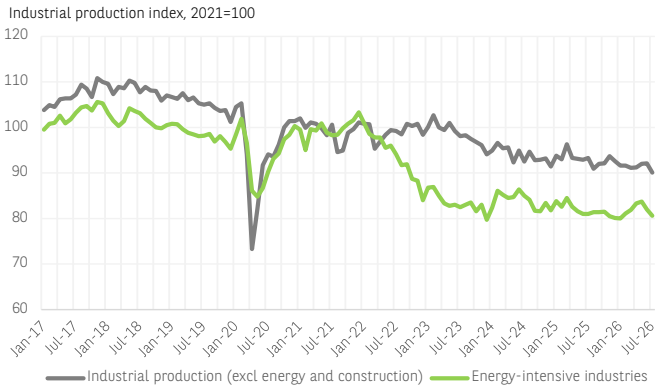


CHART 8 SOURCE: DESTATIS, BNP PARIBAS

and struggles in implementing the programme in practice within each Member State[11].

State aid is still the main vehicle for this. In response to the 2021–2022 energy crisis, the European Union temporarily relaxed the rules on this type of aid in order to support businesses facing rising energy prices. Any intervention of this kind, provided that it is selective and could affect trade within the single market, must be authorised in advance by the European Commission in principle. Since June 2025, the Clean Industrial Deal State Aid Framework (CISAF) has provided a more sustainable framework for these kinds of interventions. In particular, this framework, which runs until the end of 2030, authorises Member States to grant temporary aid to electricity-intensive businesses to offset the cost of electricity. The aid must be targeted and come with decarbonisation commitments[12]. Against the backdrop of the crisis in the Middle East, the temporary METSAF scheme, adopted in April 2026, helped (in relation to measures falling under the CISAF) to increase the proportion of additional costs offset by governments from 50% to 70%, thereby offering additional support to the most at-risk sectors.

GERMANY IS STILL SUFFERING FROM INSUFFICIENT INVESTMENT IN ITS ELECTRICITY NETWORKS, WHICH THE AID SCHEMES ARE ATTEMPTING TO OFFSET

The issue of electricity costs is a particularly hot issue in Germany, given the importance of industry to the economy (around 20% of value added, one of the highest levels in Europe) and its exposure to international competition. Furthermore, the delay in modernising electricity infrastructure (only 22.6% of the high-voltage lines planned for 2025 had actually been completed by that date) is causing bottlenecks between the production areas in the north and the consumption centres in the south. As a result, the costs of accessing services have increased fivefold over 10 years (rising from EUR 1 bn in 2014 to EUR 5.2 bn in 2024[13]). At the same time, as we have already seen, taxation and grid charges are hitting industry hard, with levies on large consumers far exceeding the European average despite some support mechanisms being put in place for the most electricity-intensive companies (such as the Bandlastprivileg measure to reduce grid tariffs). As a result, German industrial production has suffered in particular due to rising energy prices in recent years, including electricity, with electricity-intensive industries being especially affected (*Chart 8*).

In order to cope with these higher electricity costs than the European average, Germany has been using European support schemes to reduce the cost of electricity for businesses, particularly since the Merz government took office in May 2025. The main scheme — the Industriestrompreis[14], made possible by the CISAF — has been approved by the European Commission for the 2026–2028 period. While it is less beneficial than the indirect carbon cost (ICC) offsetting included in electricity prices[15], it has a broader scope, meaning that businesses ineligible for the ICC can benefit from it. Other measures complement this targeted support by addressing other components of the electricity bill. As a result, the electricity tax has been set at the European minimum for manufacturing sectors, while the gas-storage tax has been abolished. The government is also subsidising grid fees, which should halve them.

Overall, according to our estimates based on 2024 prices, the final price of electricity for these industries could fall by around a quarter (to 116 EUR/MWh for industries with annual consumption of between 70 and 150 GWh, compared with a European average of 123 EUR/MWh) (*Chart 9*). Germany's advantage over Central and Eastern European countries, where electricity costs are generally already higher, looks set to increase.

11 European Court of Auditors, 2026, "Implementation of the REPowerEU plan must be stepped up".
12 It applies to the most electricity-intensive activities, which are seeing their competitiveness threatened by international competition, and follows the 4x50 principle: 50% of electricity consumption is eligible for a subsidy; Of the eligible proportion, the subsidy may not exceed 50% of the wholesale price;
The final wholesale price paid by the company must not be less than €50/MWh;
50% of the aid received must be reinvested in decarbonisation (renewable energy generation, storage and improvements in energy efficiency); Around 90 sectors are covered by this reform, accounting for approximately 2,000 companies, including in the chemical, steel and paper industries. The scheme, which is strictly regulated, will run for just three years (from 1st January 2026 to the end of 2028) at an estimated total cost of between EUR 3 and 5 bn, financed by the special Climate and Transformation Fund.
13 Bundesnetzagentur Monitoring Report (2015) and Bundesnetzagentur Monitoring Report (2025). These costs mainly cover the costs of physical infrastructure; operational, regulatory and compliance costs; and costs associated with using the transmission and distribution network.
14 The *Industriestrompreis* (industrial electricity price) is the public subsidy mechanism aiming to reduce the price of electricity for some industries.
15 The indirect carbon cost compensation mechanism is a government support measure for electricity-intensive industries, which aims to offset the cost of carbon-emission allowances for energy producers.

However, even when taking all of the subsidies into account, the price of electricity in Germany would remain higher than the levels seen in France, Spain or Portugal (and even more so in the United States and China).

FISCAL MEASURES: LIMITED ROOM FOR MANOEUVRE

There are a number of limitations to these European schemes. For the country implementing them, lower electricity costs mean that its industry becomes more competitive against industries from other EU countries, but this does very little to address the disadvantage that it faces compared with the United States and China. Therefore, this measure will unlikely be enough to counteract the loss of market share seen vis-à-vis the United States and China. In addition, the support schemes are restrictive (conditional on investment in the low-carbon transition) and are transitional. Finally, their cost is partly passed on to the public purse or to consumers who do not receive this support (in Germany, the cost of the new energy price support measures is estimated at EUR 14.3 bn in 2026, or 0.3% of GDP[16]).

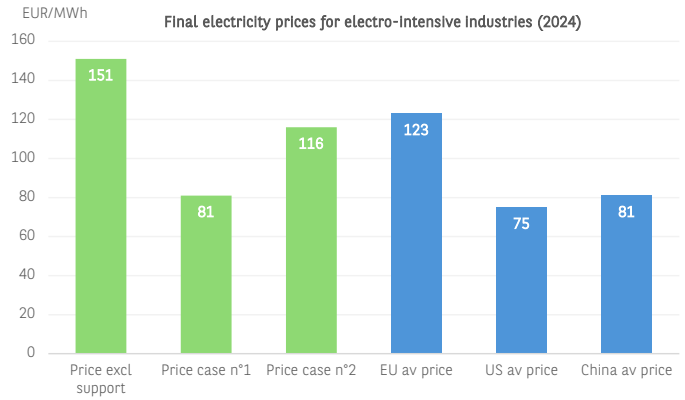
Some European manufacturers have proposed abolishing or reducing the carbon tax as a solution to getting electricity prices down quickly. In Germany, this tax accounted for 14% of the electricity price in the first half of 2025, which is higher than the EU average (11%). However, this proposal is already covered by the indirect carbon cost offsetting mechanism for electricity-intensive industries exposed to carbon allowance prices. In addition, while adjustments are being made — notably to slow the rate at which allowances are reduced — it seems highly unlikely that the scheme will be significantly challenged. It is one of the cornerstones of Europe's policy to combat climate change, which has been effective thus far, as shown by the falling greenhouse gas emissions from industry. More broadly, the gradual introduction of a carbon border adjustment mechanism (CBAM) should mitigate competition distortions caused by environmental practices[17].

IN GERMANY AND THE REST OF EUROPE, HEAVY INVESTMENT IS DRIVING UP ELECTRICITY BILLS IN THE SHORT TERM

In response to the challenge of the low-carbon transition and industrial competitiveness, European governments must strike a balance between decarbonising the electricity mix and ensuring a supply of energy that is flexible enough (particularly from controllable sources) to guarantee stable electricity systems. The integration of an increasing proportion of intermittent renewable energy sources requires massive investment in transmission and storage infrastructure (stationary batteries). This expenditure, which places a long-term burden on grid costs, reduces the scope for support schemes (particularly public ones) to address competitiveness gaps. In Germany, the cost of the transition is estimated at between EUR 122 bn, according to the government (2.8% of GDP), and EUR 200 bn, according to the DIHK[18] (4.6% of GDP), by 2030–2035. If we take the ECB's assumption[19] that governments will cover around 17% of these investments, the annual public expenditure burden of the transition in Germany would be between 0.5 and 0.8 percentage points of GDP by 2030. However, this budgetary pressure comes at the same time as three other budgetary increases: rising social expenditure linked to an ageing population; the necessary increase to defence budgets to meet NATO's target of 3.5% of GDP by 2029; and the rising cost of debt-servicing[20].

Germany has already made significant progress in maximising its use of renewable energy, putting it at the head of the pack in Europe. The country generates 20% of Europe's renewable electricity (solar and wind) and accounts for 30% of installed storage capacity (BESS)[21], far ahead of other European countries (ranking third globally). However, despite this momentum, Germany's dependence on gas has not yet decreased. As a matter of fact, it is only partially offsetting the planned closure of coal-fired power stations and the phasing out of nuclear reactors. Furthermore, the intermittent nature of solar and wind power could prolong this dependence in the short and medium term. In order to ensure that the German electricity system is flexible, new investments in gas-fired power plants, supported by public subsidies, are planned[22].

Estimated impact of Germany's measures



Notes:
The data for electricity-intensive industries are from Eurostat. Annual electricity consumption ranges from 70 to 150 GWh. Case 1: Electricity price charged to an industry benefiting from the existing indirect carbon cost compensation (ICC) mechanism and complementary support schemes. Case 2: Price charged to industries benefiting from the new Industriestrompreis measures under the European CISAF, combined with complementary support schemes.

CHART 9 SOURCE: EUROSTAT, BDEW, IEA, NDRC, BNP PARIBAS CALCULATION

16 The *Industriestrompreis* is estimated at around EUR 1.3 bn per year, the subsidy for network charges is projected to amount to EUR 6.5 bn in 2026, the abolition of the gas surcharge around EUR 3.4 bn, and the reduction in the electricity tax nearly EUR 3 bn per year. Furthermore, the ICC had cost EUR 2.3 bn in 2024 for costs incurred in 2023.
17 Jean-Luc Proutat, European Union: the carbon tax at the border in six questions, 6 October 2025.
18 German Chamber of Commerce and Industry, "Frontier economics, DIHK, 2025, New pathways for the energy transition ('Plan B')".
19 ECB, 2025, *Investing in Europe's green future. Green investment needs, outlook and obstacles to funding the gap, Occasional Paper Series*, No. 367.
20 Interest expenditure is expected to rise from 1.4% to 1.7% of GDP by 2030, due to rising refinancing rates.
21 Energy Institute, 2026, *Statistical Review Home* / Statistical Review of World Energy
22 Calls for tender are currently under way to build 12 GW of additional electricity generation capacity, with 10 GW covering gas-fired power stations (equivalent to one-third of the existing gas-fired power stations) and 2 GW potentially allocated to batteries (accounting for 80% of existing large-capacity stationary batteries intended for the wholesale market).

SUBSIDIES ARE HELPING TO ADDRESS THE EMERGENCY CAUSED BY ENERGY CRISES WITHOUT RESTORING THE INTERNATIONAL COMPETITIVENESS OF ENERGY-INTENSIVE INDUSTRIES

In response to the impact of energy crises on industrial competitiveness, European governments are finding themselves having to strike a delicate balance between providing short-term support to industries exposed to increased international competition and continuing — if not, accelerating — energy transition policies which will result in higher costs for electricity-network users for several years to come.

Beyond direct economic support measures, which are difficult to sustain long term, four priority areas are emerging:

1/ Demonstrating flexibility in implementing European directives in order to factor in national specificities and the differing timeframes between energy crises (which can lead to irreversible industrial consequences) and reforms (which have a longer-term horizon). The relaxation of the European carbon tax mechanism[23] appears to be a step towards this.

2/ Continuing efforts to decarbonise the electricity mix, which are still the main vehicle for sustainably reducing electricity prices for European industries in the medium and long term.

3/ Optimising the use of electricity networks on the demand side, in particular through smart grids, a flagship element of the Commission's COM (2026) 600 reform presented last July[24].

4/ Coordinating European electricity networks in order to harness the complementarities between national systems.

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23 In particular, the extension of the period for allocating emission allowances free of charge to sectors most exposed to international competition, and the reduction in the linear reduction factor.
24 EUR-Lex - 52026PC0600 - EN - EUR-Lex and EUR-Lex - 52026DC0595 - EN - EUR-Lex

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